MotifAE Reveals Functional Motifs from Protein Language Model: Unsupervised Discovery and Interpretability Analysis

Chao Hou[1,#], Di Liu[2], Yufeng Shen[1,2,3,#]

1 Department of Systems Biology, Columbia University Irving Medical Center, New York, NY 10032

2 Department of Biomedical Informatics, Columbia University Irving Medical Center, New York, NY 10032

3 JP Sulzberger Columbia Genome Center, Columbia University, New York, NY 10032

Corresponding author: ch3849@cumc.columbia.edu (C. H.), ys2411@cumc.columbia.edu (Y. S.)

Abstract

Protein motifs are conserved elements that mediate processes such as folding, binding, catalysis, and post-translational modifications. While motif identification is critical for protein study, experimental methods are labor-intensive, only a few hundred motifs are cataloged in databases like ELM, and existing supervised models are typically limited to predicting motifs with a specific function. Here, we present MotifAE, an unsupervised framework for discovering functional motifs from the protein language model ESM2, which captures evolutionary-scale sequence regularities. MotifAE is based on the sparse autoencoder (SAE), an encoder-decoder architecture that projects ESM2 embeddings into a sparse latent space, with an additional local similarity loss that encourages coherent latent feature activations. When benchmarked against known ELM motifs, MotifAE achieves a median AUROC of 0.88, outperforming the standard SAE (0.80). We also calculated Position-specific scoring matrices (PSSMs) for MotifAE features and found that features with similar decoder weights share similar PSSMs. Furthermore, by aligning MotifAE features with experimental data through gated feature selection, we identified features associated with specific properties such as folding stability. Steering these features enabled designing proteins with enhanced stability, as evaluated *in silico*. Overall, MotifAE provides a general framework for systematic motif discovery and interpretation, with the potential to advance protein function analysis, mutation effect interpretation, and rational protein engineering.

Introduction

Proteins mediate essential biological processes, such as catalysis, immune defense, signal transduction, and molecular transport. Their functional regions, such as catalytic centers, binding interfaces, and post-translational modification sites, exhibit conserved sequence patterns, commonly referred to as motifs. Systematic discovery of functional motifs is crucial for advancing our understanding of protein function and has broad applications in gene annotation, mutation effect interpretation, and protein engineering.

Experimental Identification of functional motifs requires first locating protein regions that share the same function, then aligning these regions to get the motif sequence pattern, typically represented as a regular expression or a position-specific scoring matrix (PSSM)[1]. This process is labor-intensive and time-consuming. As of 2025, only 353 motifs have been curated in the Eukaryotic Linear Motif (ELM) database[2]. High-throughput screens have been developed to map functional regions[3–6], but these screens remain limited in scale and are tailored to specific and readily measurable functions, such as protein abundance or binding to a specific partner, making them unsuitable for systematically identifying the full spectrum of functional motifs. To complement experimental efforts, machine learning methods have also been developed[7–9], often trained on curated resources like ELM or datasets from high-throughput screens. However, these supervised models are typically tailored to specific functions and inherently biased toward their training datasets, limiting their generalizability.

In recent years, deep learning has driven remarkable advances in modeling protein sequence[10], structure[11], and function[12]. A major development is protein language models (pLMs), such as ESM2[10], which are self-supervised models trained on large-scale protein databases using masked or next-token prediction. Through this pre-training, pLMs implicitly learn conserved sequence patterns at the evolutionary scale. For example, Zhang et al.[13] demonstrated that ESM2 predicts protein structure by storing pairwise contact motifs, Vig et al.[14] found that pLMs' attentions capture target binding sites, Zhang et al.[15] showed that mutational constraints predicted by ESM2 are predictive of conserved sequence motifs. Moreover, numerous studies have leveraged pLM embeddings to predict functional regions such as signal peptides, transit peptides, and post-translational modification sites[8]. Collectively, these works demonstrate that pLMs encode rich information about functional motifs within their parameters. However, because pLMs operate as black boxes, the motif information they encode is not directly accessible.

Various attempts have been made to extract interpretable features from language models. Among them, sparse autoencoders (SAEs) have shown strong potential for interpretability[16]. The core assumption of SAEs is that the model embedding can be expressed as a linear combination of different features[17], with each feature ideally corresponding to an interpretable factor. To achieve this, SAE uses an encoder to project embeddings into a higher-dimensional, sparse latent space, and uses a decoder to reconstruct the embeddings from this space. Hereafter, we refer to each neuron in the SAE latent space as a feature, and neurons with positive values are activated. During SAE training, a reconstruction loss was used to ensure that the embeddings are accurately reconstructed, and a sparsity constraint was used to encourage sparse feature activations. Typically, only tens of features are active per residue, far fewer than the original embedding dimension, which often exceeds a thousand. SAEs have been successfully applied in large language models (LLMs) to find interpretable semantic features[16,17]. Adapting the training and interpretation strategies developed for LLM SAEs, similar approaches in biological sequence models have identified known functional elements, including DNA features such as exons, introns, and transcription factor binding sites[18], as well as protein features such as binding motifs and structural domains[19–21].

Here, we developed MotifAE, introducing methodological advances in both SAE model training and interpretation to make them better suited for biological sequences and compatible with experimental data. We incorporated an additional local similarity loss during training, encouraging MotifAE's latent features to capture the sequential nature of motifs. Compared to standard SAEs, MotifAE markedly improves the identification of functional motifs across diverse benchmarks. Building on this capability, we further align MotifAE features with experimental protein fitness data using a gated feature selection approach, which enables the identification of associated features and enhances performance in fitness prediction. Moreover, steering these selected features allows protein design with desired properties, as evaluated *in silico*. Overall, MotifAE provides a systematic framework for motif discovery from pLMs and annotation using experimental data, facilitating deeper insights into protein function, and offering potential applications in gene annotation, mutation effect interpretation, and protein engineering.

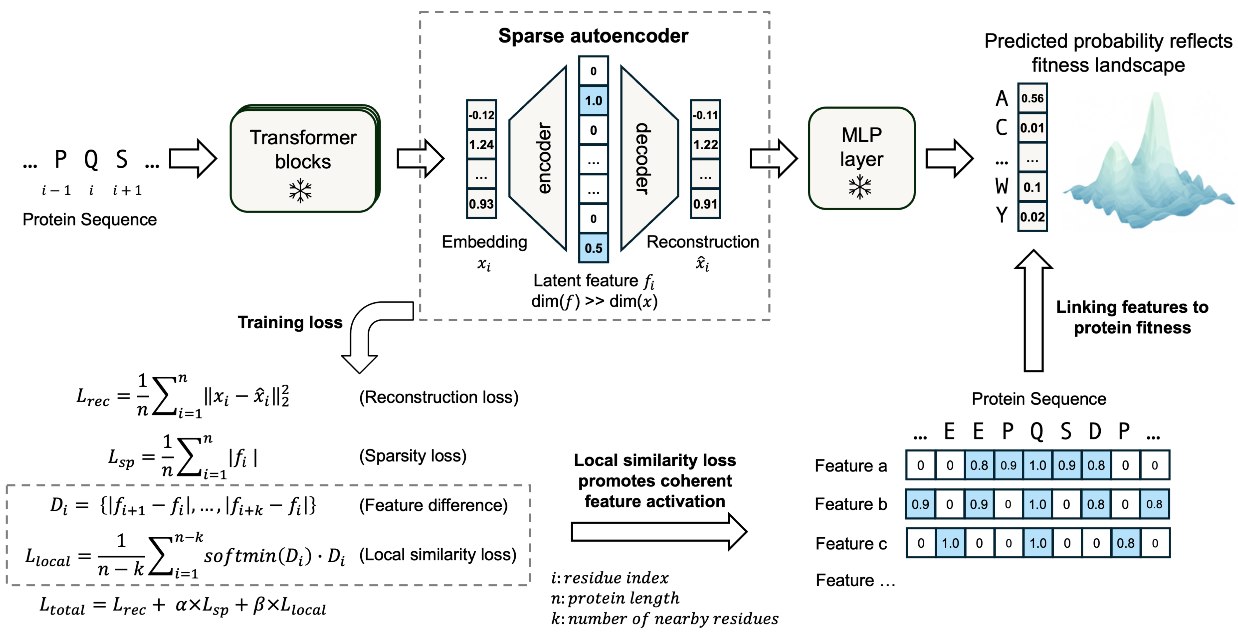

Figure 1. The MotifAE framework.

MotifAE uses the same architecture as the sparse autoencoder (SAE) but is trained with an additional local similarity loss. MotifAE projects ESM2 embeddings (dimension 1,280) into a high-dimensional latent space (dimension 40,960) using an encoder, and then reconstructs the embeddings through a decoder. The reconstructed embeddings can be mapped to amino acid probabilities via the fixed ESM2 MLP layer, enabling prediction of protein fitness landscape. MotifAE is trained with three objectives: a reconstruction loss, a sparsity loss, and a local similarity loss, the latter encourages locally coherent latent feature activations.

Results

MotifAE is a sparse autoencoder with coherent latent feature activation.

We developed MotifAE, an adaptation of the sparse autoencoder (SAE) for discovering functional motifs from pLMs. A standard SAE is trained with a reconstruction loss and a sparsity constraint (**Figure 1**), ensuring that the original embeddings are accurately reconstructed while enforcing sparse activation of latent features. To enhance biological relevance, we introduced a local similarity loss (see Methods) that encourages MotifAE features to activate coherently along the sequence. This design is motivated by the fact that protein motifs often involve local contiguous residues[2], and basic structural elements also exhibit local patterns, such as alternating contacts in β-strands and periodic interactions every three or four residues in α-helices. The local similarity loss requires that at least one neighboring residue within a defined window exhibits activation features similar to the current residue (**Figure 1**; see Methods), thereby promoting local coherence in feature activation. A window size of three residues was used in this study; since the local similarity loss is computed over the entire sequence, it can capture relationships extending beyond the three-residue window.

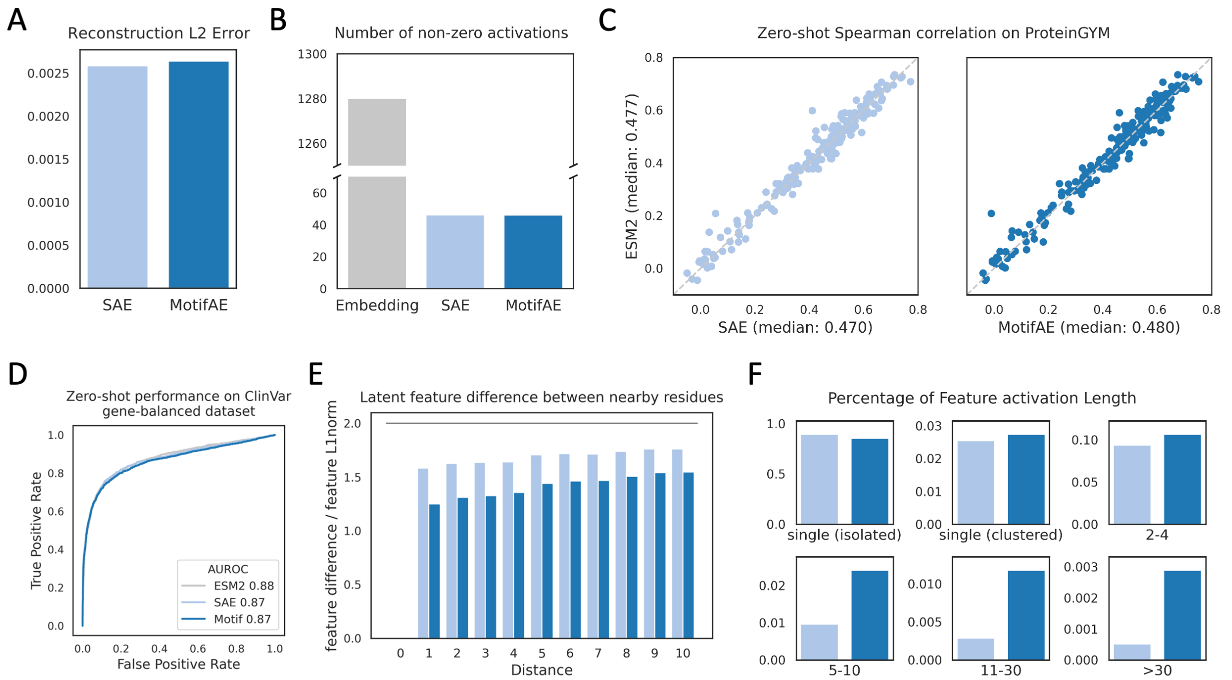

Figure 2. Comparison of MotifAE and SAE in terms of reconstruction error, sparsity, fitness prediction, and activation length distribution.

(**A**) Embedding reconstruction error on the evaluation set (see Methods). (**B**) Mean number of non-zero neurons per residue. The ESM2 embedding has 1280 non-zero neurons, while both SAE and MotifAE have ~46 non-zero (positive) latent features per residue. (**C**) Zero-shot performance (spearman correlation) on ProteinGYM DMS datasets, only single substitution mutations were evaluated. Each dot represents a DMS dataset. Wild-type marginal log-likelihood ratio (LLR) was used to estimate mutation effects. (**D**) Zero-shot performance on classifying pathogenic and benign missense mutations in the ClinVar gene-balanced dataset (see Methods). (**E**) Latent feature difference between nearby residues with different distances, the value of y-axis is 0 for identical activations and ~2 for random sparse activations. (**F**) Length distribution of activations from all features. For each feature, A single (clustered) activation is defined as an activated single residue with other activated residues within two amino acids along the sequence. In all plots, light blue denotes SAE, and dark blue denotes MotifAE.

We trained MotifAE on the final-layer embeddings from ESM2-650M, which has shown strong performance across diverse downstream tasks. ESM2-650M captures the protein fitness landscape by being trained on evolutionary-scale sequences and is widely used to predict mutation effects[22–24]. We used the last-layer embeddings because they are directly used to predict amino acid probability which reflects the protein fitness landscape[22,23], allowing us to investigate the relationship between latent features and fitness. To ensure that MotifAE captures representative protein features, training was performed on 2.3 million representative proteins obtained from structure-based clustering of the AlphaFold2 structure database[25]. The dimension of the latent space and weights for different losses were determined by jointly considering reconstruction error, latent sparsity, and the protein fitness prediction performance (**Figure S1**; see Methods). Based on these criteria, we selected a hidden dimension of 40,960, corresponding to a 32-fold expansion over the ESM2 embedding dimension. While only embeddings from unmasked sequences were used for model training, MotifAE can reconstruct embeddings of masked residues, and the reconstructed embeddings preserve essential information for predicting the masked amino acids when processed with the fixed ESM2 MLP layer (**Figure S1C**). We note that the reconstruction of ESM2 embeddings is highly fragile. Although the reconstruction